

Residual-Mass Transform Order for LRU Caching and the Coupon Collector

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Abstract

For the independent reference model with a positive popularity vector $p \in \Delta_N^\circ$, let $S_C(p)$ be the first C distinct items in size-biased order, equivalently the stationary LRU cache set of capacity C . Put

$$M_C(p) = p(S_C(p)), \quad Q_C(p) = 1 - M_C(p).$$

Thus M_C is the random cached mass and Q_C is the random residual miss mass. We prove a finite-dimensional radial transform-order theorem. Along every nonconstant ray $p_\theta = u + \theta(p - u)$ from the uniform law, $M_C(p_\theta)$ is increasing in increasing-convex order. Equivalently, for every decreasing convex cost ψ ,

$$\theta \mapsto \mathbb{E}\psi(Q_C(p_\theta))$$

is nondecreasing, and is strict for the natural strictly monotone costs. The proof is a pathwise crossing argument using a common Gumbel representation of the Plackett–Luce size-biased order: higher-than-uniform coordinates can only overtake lower-than-uniform coordinates as θ increases.

The result simultaneously strengthens radial LRU extremality and gives a new exact coupon-collector transfer principle. In LRU, it implies monotonicity not only of the mean hit and miss rates but also of no-miss survival probabilities. In coupon collecting, it implies that every one-step discovery increment $T_{C+1} - T_C$ is radially increasing in stochastic order, and that

$$\theta \mapsto \mathbb{E}Q_C(p_\theta)^{-s}$$

is strictly increasing for all $s > 0$. The exact residual-mass identities

$$\text{MR}_{LRU}(C; p) = \mathbb{E}Q_C(p), \quad \mathbb{E}_p[T_{C+1} - T_C] = \mathbb{E}Q_C(p)^{-1}$$

therefore become two projections of a single ordered residual-hazard law. This clarifies and strengthens the asymptotic LRU–coupon-collector relation of Berthet.

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1 Introduction

The independent reference model (IRM) is the standard model in which successive requests are independent with common law

$$p = (p_1, \dots, p_N) \in \Delta_N^\circ, \quad p_i > 0, \quad \sum_i p_i = 1.$$

The same popularity vector governs two classical objects.

First, for an LRU cache of capacity C , the stationary cache set is the set of the first C distinct items encountered when scanning the IRM sequence backwards from stationarity. Equivalently, it is the set of the first C items in a size-biased permutation with weights p . Second, in the coupon collector, the set of the first C distinct coupons seen in a forward run has the same size-biased distribution. Thus both models share the same random prefix set S_C .

The central random variable of this paper is the residual mass

$$Q_C = 1 - p(S_C).$$

Its complement

$$M_C = p(S_C)$$

is the random mass already represented in the size-biased prefix. In LRU language, M_C is the conditional hit probability given the current cache set and Q_C is the conditional miss probability. In coupon-collector language, Q_C is the conditional success probability for discovering a new coupon after C distinct coupons have already been collected.

The exact identities

$$\text{MR}_{LRU}(C; p) = \mathbb{E}Q_C(p), \quad \mathbb{E}_p[T_{C+1} - T_C] = \mathbb{E}Q_C(p)^{-1} \tag{1}$$

are elementary, but they are structurally revealing. They show that the LRU miss rate and the coupon-collector waiting increment are the positive and reciprocal first moments of the same random residual mass. The product relation

$$\text{MR}_{LRU}(C; p) \mathbb{E}_p[T_{C+1} - T_C] = \mathbb{E}Q_C \mathbb{E}Q_C^{-1} \geq 1$$

is therefore a Jensen gap. Equality holds at uniform popularity; away from uniform it measures dispersion of the residual mass.

Berthet [3] derived and emphasized an asymptotic relation of the form

$$\text{MR}_{LRU}(C; p) (\mathbb{E}T_{C+1} - \mathbb{E}T_C) \approx 1$$

between LRU miss rates and coupon-collector waiting-time increments. The identities above show that the approximation is best understood as a concentration statement for Q_C , not as an exact reciprocal relation between the two expectations. The present paper strengthens this viewpoint by proving a transform-order theorem for Q_C along rays from uniform.

Let

$$u = (1/N, \dots, 1/N), \quad p_\theta = u + \theta(p - u), \quad 0 \leq \theta \leq 1.$$

Our main result says that, for every $1 \leq C \leq N - 1$, the cached mass $M_C(p_\theta)$ is increasing in increasing-convex order. Equivalently, for every decreasing convex function ψ ,

$$\theta \mapsto \mathbb{E}\psi(Q_C(p_\theta))$$

is nondecreasing. This single statement gives, among other consequences,

$$\mathbb{E}Q_C(p_\theta) \downarrow, \quad \mathbb{E}e^{-\lambda Q_C(p_\theta)} \uparrow, \quad \mathbb{E}Q_C(p_\theta)^{-s} \uparrow, \quad \mathbb{E}Q_C(p_\theta)^r \downarrow \quad (0 < r \leq 1).$$

Thus the LRU miss cost q and the coupon-collector reciprocal cost q^{-1} are two special costs in a larger ordered family.

Relation to earlier LRU radial extremality

A previous paper [13] proved that the exact stationary LRU hit rate $\text{Hit}_C(p) = \mathbb{E}M_C(p)$ is strictly increasing along every nonconstant ray from uniform. That result used the exponential-age representation of LRU and produced an explicit positive pair-square radial derivative formula. It also established the radial restriction of the Fill–Holst conjecture for move-to-front search-cost tails and explained why full Schur-convexity fails.

The present theorem strictly extends the conclusion of [13]: the hit-rate monotonicity is the linear case $\phi(x) = x$ of increasing-convex order for M_C . However, the earlier positive-kernel proof still has independent value. It gives an explicit local derivative certificate and computable pair kernels; it also isolates the finite-dimensional algebraic positivity that survives on rays even though arbitrary majorization directions can fail. The Gumbel crossing method developed here gives a shorter and stronger global order theorem, while the pair-kernel proof remains useful for sensitivity analysis, local certificates, and transfer to models where a common-Gumbel ranking representation is unavailable.

Related work

Flajolet, Gardy and Thimonier [9] gave a unified analytic framework for the birthday paradox, coupon collectors, caching algorithms and self-organizing search under general nonuniform distributions. Exact and asymptotic LRU analysis under the IRM goes back at least to King, Fagin and Fagin–Price [12, 6, 7], with later asymptotic work on move-to-front and LRU fault probabilities by Jelenkovic [11]. Che, Tung and Wang [5] introduced the characteristic-time approximation that is now a standard engineering approximation for LRU. Boneh and Hofri [4] surveyed coupon-collector formulae and computational methods. Anceaume, Busnel, Schulte-Geers and Sericola [2] proved stochastic optimization results for generalized coupon collection, including almost-uniform minimization for the time to collect a fixed number of distinct coupons.

The present contribution is different in focus: it identifies an exact finite- N residual-hazard law common to LRU and the coupon collector, and proves a radial transform order for that law.

2 Size-biased order, LRU, and coupon collection

2.1 Size-biased permutations

Let $p \in \Delta_N^\circ$. A size-biased permutation with weights p is the random permutation $(\Pi_1, \dots, \Pi_N)$ satisfying

$$\mathbb{P}(\Pi_1 = i_1, \dots, \Pi_N = i_N) = \prod_{r=1}^N \frac{p_{i_r}}{1 - \sum_{s < r} p_{i_s}}, \quad (2)$$

for every ordering $(i_1, \dots, i_N)$ of $[N]$. This is the Plackett–Luce distribution.

For $0 \leq C \leq N$, put

$$S_C = \{\Pi_1, \dots, \Pi_C\},$$

with $S_0 = \emptyset$ and $S_N = [N]$. Define

$$M_C = p(S_C) = \sum_{i \in S_C} p_i, \quad Q_C = 1 - M_C.$$

The nontrivial capacities are $1 \leq C \leq N - 1$. At $C = 0$, $Q_0 = 1$; at $C = N$, $Q_N = 0$.

Lemma 2.1 (Exponential and Gumbel representations). *Let $E_1, \dots, E_N$ be independent exponential random variables with rate one. Then the increasing order of*

$$A_i = \frac{E_i}{p_i}$$

has the size-biased law (2). Equivalently, if

$$G_i = -\log E_i,$$

then G_i are independent standard Gumbel variables, and the decreasing order of

$$Y_i = G_i + \log p_i$$

has the size-biased law.

Proof. The variables A_i are independent exponentials with rates p_i . The first clock to ring has label i with probability $p_i / \sum_j p_j = p_i$. Conditional on the first label, the residual clocks of the unchosen items are independent exponentials with their original rates, by memorylessness. Iterating gives (2). Since increasing order of E_i/p_i is the same as decreasing order of $-\log E_i + \log p_i$, the Gumbel form follows. $\square$

2.2 LRU cache sets

Under the IRM, scan the stationary request sequence backwards from a reference time. The first distinct item encountered is distributed as p . Given that a set A has already been encountered, the next new item is $i \notin A$ with probability

$$\frac{p_i}{1 - p(A)}.$$

Therefore the backward order of distinct items is a size-biased permutation. The LRU cache of capacity C contains exactly the first C distinct items in this backward scan. Thus its stationary cache set has the same law as S_C .

Consequently,

$$\text{Hit}_{LRU}(C; p) = \mathbb{E}M_C(p), \quad \text{MR}_{LRU}(C; p) = \mathbb{E}Q_C(p). \quad (3)$$

Moreover, conditional on the stationary cache set, the next request hits with probability M_C and misses with probability Q_C . Hence, in discrete time,

$$\mathbb{P}(\text{the next } n \text{ requests are all LRU hits}) = \mathbb{E}M_C^n, \quad n \geq 0. \quad (4)$$

In the rate-one Poissonized IRM,

$$\mathbb{P}(\text{no LRU miss before time } t) = \mathbb{E}e^{-tQ_C}. \quad (5)$$

2.3 Coupon-collector increments

Let T_C be the number of draws needed to have seen C distinct coupon types, with $T_0 = 0$. The set of the first C collected types is distributed as S_C . Conditional on S_C , the probability that the next draw is a new coupon is Q_C . Therefore

$$Y_C := T_{C+1} - T_C$$

is conditionally geometric on $\{1, 2, \dots\}$ with success probability Q_C :

$$\mathbb{P}(Y_C > n \mid S_C) = (1 - Q_C)^n = M_C^n, \quad n \geq 0. \quad (6)$$

Thus

$$\mathbb{E}[Y_C] = \mathbb{E}Q_C^{-1}. \quad (7)$$

In the Poissonized coupon collector, the corresponding waiting time from C to $C + 1$ distinct coupons is conditionally exponential with rate Q_C , and therefore has survival

$$\mathbb{P}(Y_C^{\text{cont}} > t) = \mathbb{E}e^{-tQ_C}. \quad (8)$$

Proposition 2.2 (Exact residual-mass identities). *For every $p \in \Delta_N^\circ$ and $0 \leq C < N$,*

$$\text{MR}_{LRU}(C; p) = \mathbb{E}Q_C(p), \quad \mathbb{E}_p[T_{C+1} - T_C] = \mathbb{E}Q_C(p)^{-1}. \quad (9)$$

Consequently, for $1 \leq C < N$,

$$\text{MR}_{LRU}(C; p) \mathbb{E}_p[T_{C+1} - T_C] = \mathbb{E}Q_C \mathbb{E}Q_C^{-1} \geq 1. \quad (10)$$

Equality holds if and only if p is uniform.

Proof. The identities are (3) and (7). The inequality is Cauchy's inequality, or Jensen's inequality applied to $q \mapsto q^{-1}$:

$$\mathbb{E}Q_C^{-1} \geq (\mathbb{E}Q_C)^{-1}.$$

Equality holds if and only if Q_C is almost surely constant. Since $p_i > 0$, every C -subset occurs as S_C with positive probability. Thus Q_C is constant if and only if $p(A)$ is the same for every C -subset A . Comparing two C -subsets that differ by replacing i with j gives $p_i = p_j$ for all i, j . Hence $p = u$. $\square$

3 The crossing method

Fix a positive nonuniform vector $p \in \Delta_N^\circ$ and write

$$h_i = p_i - \frac{1}{N}, \quad p_i(\theta) = \frac{1}{N} + \theta h_i, \quad 0 \leq \theta \leq 1.$$

Then $\sum_i h_i = 0$ and $p_i(\theta) > 0$ for $0 \leq \theta \leq 1$.

Use a single family of independent standard Gumbel variables $G_1, \dots, G_N$ for every θ , and define

$$Y_i(\theta) = G_i + \log p_i(\theta). \quad (11)$$

Let $S_C(\theta)$ be the set of indices corresponding to the C largest values of $Y_i(\theta)$. By Lemma 2.1, $S_C(\theta)$ has the size-biased prefix law with weights p_θ .

Define the h -mass of the prefix by

$$H_C(\theta) = \sum_{i \in S_C(\theta)} h_i. \quad (12)$$

Then

$$M_C(\theta) = p_\theta(S_C(\theta)) = \frac{C}{N} + \theta H_C(\theta). \quad (13)$$

Lemma 3.1 (One-way crossings). *For $i \neq j$,*

$$\frac{d}{d\theta}(Y_i(\theta) - Y_j(\theta)) = \frac{(1/N)(h_i - h_j)}{p_i(\theta)p_j(\theta)}. \quad (14)$$

Consequently, if $h_i > h_j$, then item i can overtake item j as θ increases, but item j cannot overtake item i .

Proof. Since

$$\frac{d}{d\theta} \log p_i(\theta) = \frac{h_i}{p_i(\theta)},$$

we have

$$\begin{aligned} \frac{d}{d\theta}(Y_i - Y_j) &= \frac{h_i}{p_i(\theta)} - \frac{h_j}{p_j(\theta)} \\ &= \frac{h_i p_j(\theta) - h_j p_i(\theta)}{p_i(\theta)p_j(\theta)} \\ &= \frac{(1/N)(h_i - h_j)}{p_i(\theta)p_j(\theta)}. \end{aligned}$$

The denominator is positive. Thus the sign is the sign of $h_i - h_j$. $\square$

Lemma 3.2 (Pathwise monotonicity of prefix h -mass). *For every fixed realization of the Gumbel variables outside a null set, the function*

$$\theta \mapsto H_C(\theta)$$

is nondecreasing on $[0, 1]$.

Proof. Almost surely there are no triple ties and no pair is tied at a prescribed endpoint. For each pair i, j , Lemma 3.1 shows that their relative order can change at most once, and when it changes the item with larger h moves upward past the item with smaller h .

The top- C set changes only when a crossing occurs across the boundary between ranks C and $C + 1$. At such a crossing, the entering item has larger h than the leaving item. Therefore the sum of h_i over the top- C set increases. Crossings entirely inside the top C or entirely outside it leave the set, and hence H_C , unchanged. Since there are only finitely many pairs, $H_C(\theta)$ is a step function with nonnegative jumps. $\square$

The next elementary lemma is the analytic step that converts pathwise monotonicity of H_C into increasing-convex order for M_C .

Lemma 3.3 (A scaling lemma for increasing convex costs). *Let X be a bounded real random variable with $\mathbb{E}X \geq 0$. Fix $c \in \mathbb{R}$ and an interval I containing $c + aX$ for all $a \in [a_0, a_1]$, where $0 \leq a_0 \leq a_1$. If $\phi : I \rightarrow \mathbb{R}$ is increasing and convex, then*

$$a \mapsto \mathbb{E}\phi(c + aX)$$

is nondecreasing on $[a_0, a_1]$.

Proof. On a compact interval, every increasing convex function admits the representation

$$\phi(x) = \alpha + \beta x + \int (x - t)_+ \mu(dt),$$

where $\beta \geq 0$ and μ is a finite nonnegative measure. It is therefore enough to check the affine function x and the hinge functions $(x - t)_+$.

For the affine function, monotonicity follows from $\mathbb{E}X \geq 0$. For a hinge, put

$$F_t(a) = \mathbb{E}(c + aX - t)_+.$$

For $a > 0$, the right derivative is

$$F'_t(a+) = \mathbb{E}[X \mathbf{1}_{\{X > (t-c)/a\}}].$$

For any threshold b , the quantity $\mathbb{E}[X \mathbf{1}_{\{X > b\}}]$ is nonnegative. If $b \geq 0$ this is immediate. If $b < 0$, then

$$\mathbb{E}[X \mathbf{1}_{\{X > b\}}] = \mathbb{E}X - \mathbb{E}[X \mathbf{1}_{\{X \leq b\}}] \geq 0,$$

because $\mathbb{E}X \geq 0$ and $X \mathbf{1}_{\{X \leq b\}} \leq 0$. Thus F_t is nondecreasing. The case $a = 0$ follows by right continuity. $\square$

4 The transform-order theorem

Theorem 4.1 (Radial increasing-convex order for cached mass). *Let $N \geq 2$, let $1 \leq C \leq N - 1$, and let $p \in \Delta_N^\circ$ be nonuniform. Put*

$$p_\theta = u + \theta(p - u), \quad 0 \leq \theta \leq 1.$$

Then for every increasing convex function ϕ on $[0, 1]$, the map

$$\theta \mapsto \mathbb{E}\phi(M_C(p_\theta))$$

is nondecreasing. More explicitly, if $0 \leq \theta_1 < \theta_2 \leq 1$, then

$$\mathbb{E}\phi(M_C(p_{\theta_2})) \geq \mathbb{E}\phi(M_C(p_{\theta_1})). \quad (15)$$

For every strictly increasing convex ϕ , and also for the affine function $\phi(x) = x$, the inequality is strict.

Proof. Use the common-Gumbel construction from Section 3. Put

$$c = \frac{C}{N}, \quad H_r = H_C(\theta_r), \quad r = 1, 2.$$

By Lemma 3.2, $H_2 \geq H_1$ pathwise. Therefore, since ϕ is increasing,

$$\mathbb{E}\phi(c + \theta_2 H_2) \geq \mathbb{E}\phi(c + \theta_2 H_1). \quad (16)$$

It remains to compare $\theta_2 H_1$ and $\theta_1 H_1$. At $\theta = 0$, the Gumbel order is a uniformly random permutation, so $S_C(0)$ is a uniformly random C -subset. Hence

$$\mathbb{E}H_C(0) = \frac{C}{N} \sum_i h_i = 0.$$

By Lemma 3.2, $H_1 \geq H_C(0)$ pathwise, and therefore

$$\mathbb{E}H_1 \geq 0.$$

Applying Lemma 3.3 with $X = H_1$, $a_0 = \theta_1$, and $a_1 = \theta_2$ gives

$$\mathbb{E}\phi(c + \theta_2 H_1) \geq \mathbb{E}\phi(c + \theta_1 H_1). \quad (17)$$

Combining (16) and (17), and using (13), proves (15).

For strictness, choose i, j with $h_i > h_j$. Given any interval (θ_1, θ_2) , there is positive probability that $Y_i(\theta) - Y_j(\theta)$ crosses zero inside that interval. Indeed the required condition is that $G_i - G_j$ lies in a nonempty open interval, and $G_i - G_j$ has a positive density. On a further positive-probability event, exactly $C - 1$ of the remaining Gumbels are so large that they stay above both i and j throughout $[\theta_1, \theta_2]$, and all remaining Gumbels are so small that they stay below both. On this event the crossing changes the membership of the top- C set, replacing j by i , and hence H_C increases by $h_i - h_j > 0$.

Thus $\mathbb{P}(H_2 > H_1) > 0$. For strictly increasing ϕ , the first comparison (16) is strict. For $\phi(x) = x$, strictness follows from

$$\mathbb{E}M_C(\theta_2) - \mathbb{E}M_C(\theta_1) = \theta_2 \mathbb{E}(H_2 - H_1) + (\theta_2 - \theta_1) \mathbb{E}H_1 > 0.$$

This completes the proof. $\square$

Corollary 4.2 (Decreasing convex order for residual mass). *Under the hypotheses of Theorem 4.1, for every decreasing convex function ψ on $(0, 1]$,*

$$\theta \mapsto \mathbb{E}\psi(Q_C(p_\theta)) \quad (18)$$

is nondecreasing whenever the expectation is finite. It is strict for the strictly decreasing costs considered below.

Proof. Put $\phi(m) = \psi(1 - m)$. If ψ is decreasing and convex, then ϕ is increasing and convex. Since $Q_C = 1 - M_C$, Theorem 4.1 gives the claim. $\square$

5 Consequences for LRU

5.1 Hit rate, miss rate, and stack-depth tails

Taking $\phi(m) = m$ in Theorem 4.1 gives the radial LRU theorem of [13].

Corollary 5.1 (LRU hit and miss rates). *For $1 \leq C \leq N - 1$ and nonuniform $p \in \Delta_N^\circ$,*

$$\theta \mapsto \text{Hit}_{LRU}(C; p_\theta) = \mathbb{E}M_C(p_\theta)$$

is strictly increasing on $[0, 1]$, and

$$\theta \mapsto \text{MR}_{LRU}(C; p_\theta) = \mathbb{E}Q_C(p_\theta)$$

is strictly decreasing on $[0, 1]$.

Equivalently, for the move-to-front search cost R in stationarity,

$$\mathbb{P}(R > C) = \text{MR}_{LRU}(C; p).$$

Therefore the stationary move-to-front search cost decreases in usual stochastic order along every nonconstant ray away from uniform. This is the radial restriction of the Fill–Holst search-cost tail conjecture established in [13]; the present argument obtains it as the linear shadow of a stronger transform order.

5.2 No-miss survival and hit-run lengths

Taking $\phi(m) = m^n$ gives a stronger LRU statement.

Corollary 5.2 (Discrete no-miss survival). *For every integer $n \geq 1$,*

$$\theta \mapsto \mathbb{E}M_C(p_\theta)^n$$

is strictly increasing. Equivalently, starting from a stationary LRU cache of capacity C , the probability that the next n requests are all hits is strictly increasing along every nonconstant ray away from uniform.

Proof. The function $m \mapsto m^n$ is increasing and convex on $[0, 1]$. Use Theorem 4.1 and identity (4). $\square$

Taking $\psi(q) = e^{-tq}$ gives the continuous-time version.

Corollary 5.3 (Continuous no-miss survival). *For every $t > 0$,*

$$\theta \mapsto \mathbb{E}e^{-tQ_C(p_\theta)}$$

is strictly increasing. Equivalently, in the rate-one Poissonized IRM, the probability that no LRU miss occurs before time t is strictly increasing along every nonconstant ray away from uniform.

Proof. The function $q \mapsto e^{-tq}$ is decreasing and convex on $(0, 1]$. Apply Corollary 4.2 and use (5). $\square$

6 Consequences for coupon collecting

6.1 One-step stochastic monotonicity

The coupon-collector increment $Y_C = T_{C+1} - T_C$ is conditionally geometric with success probability Q_C . Its survival probabilities are $\mathbb{E}M_C^n$. Corollary 5.2 therefore gives an exact one-step stochastic order.

Theorem 6.1 (Radial stochastic monotonicity of discovery increments). *For $0 \leq C < N$ and nonuniform $p \in \Delta_N^\circ$, the coupon-collector increment*

$$T_{C+1} - T_C$$

is stochastically increasing along the ray $p_\theta = u + \theta(p - u)$. More precisely, for every $n \geq 0$,

$$\theta \mapsto \mathbb{P}_{p_\theta}(T_{C+1} - T_C > n)$$

is nondecreasing, and it is strict for $n \geq 1$ when $1 \leq C \leq N - 1$.

Proof. For $n = 0$ the survival probability is one. For $n \geq 1$, equation (6) gives

$$\mathbb{P}(T_{C+1} - T_C > n) = \mathbb{E}M_C^n,$$

and Corollary 5.2 applies. The case $C = 0$ is deterministic with $Q_0 = 1$ and $M_0 = 0$. $\square$

In continuous time, the same statement follows from Corollary 5.3 and (8).

6.2 Reciprocal and negative moments

Corollary 6.2 (Negative residual moments and CCP increments). *For every $s > 0$,*

$$\theta \mapsto \mathbb{E}Q_C(p_\theta)^{-s}$$

is strictly increasing for $1 \leq C \leq N - 1$. In particular,

$$\theta \mapsto \mathbb{E}_{p_\theta}[T_{C+1} - T_C] = \mathbb{E}Q_C(p_\theta)^{-1} \quad (19)$$

is strictly increasing.

Proof. For $s > 0$, the function $q \mapsto q^{-s}$ is decreasing and convex on $(0, 1]$. Apply Corollary 4.2. The identity (19) is (7). $\square$

Summing (19) over $C = 0, \dots, j - 1$ recovers radial monotonicity of the expected time to collect j distinct coupons:

$$\theta \mapsto \mathbb{E}_{p_\theta}T_j$$

is strictly increasing for $2 \leq j \leq N$. The theorem is stronger than this expectation statement because it gives stochastic monotonicity of every one-step increment.

7 Residual-mass cost classes

Corollary 4.2 can be restated in cost language. If f is increasing and concave on $[0, 1]$, then $-f$ is decreasing and convex, so

$$\theta \mapsto \mathbb{E}f(Q_C(p_\theta))$$

is nonincreasing. If g is completely monotone, then g is decreasing and convex, so

$$\theta \mapsto \mathbb{E}g(Q_C(p_\theta))$$

is nondecreasing.

Corollary 7.1 (Fractional positive moments). *For every $0 < r \leq 1$,*

$$\theta \mapsto \mathbb{E}Q_C(p_\theta)^r$$

is strictly decreasing for $1 \leq C \leq N - 1$.

Proof. For $0 < r < 1$, the function $q \mapsto q^r$ is increasing and concave, equivalently $q \mapsto -q^r$ is decreasing and convex. The case $r = 1$ is Corollary 5.1. $\square$

Corollary 7.2 (Completely monotone residual costs). *If g is completely monotone on $(0, 1]$ and finite on the range of Q_C , then*

$$\theta \mapsto \mathbb{E}g(Q_C(p_\theta))$$

is nondecreasing. This includes $g(q) = e^{-\lambda q}$ and $g(q) = q^{-s}$.

Proof. A completely monotone function satisfies $g' \leq 0$ and $g'' \geq 0$, hence is decreasing and convex. Apply Corollary 4.2. $\square$

7.1 Boundaries of the theorem

The theorem deliberately does not assert monotonicity of all positive moments or centered moments of Q_C .

Example 7.3 (High positive moments can fail). *For $C = 1$,*

$$Q_1 = 1 - p_I, \quad \mathbb{P}(I = i) = p_i,$$

and hence

$$\mathbb{E}Q_1^r = \sum_i p_i(1 - p_i)^r.$$

With $N = 3$, $r = 10$, and $p = (0.98, 0.01, 0.01)$,

$$\mathbb{E}_p Q_1^{10} = 0.98(0.02)^{10} + 0.02(0.99)^{10} \approx 0.0180876,$$

whereas under uniform popularity

$$\mathbb{E}_u Q_1^{10} = (2/3)^{10} \approx 0.0173415.$$

Thus uniform does not maximize high positive residual moments.

Example 7.4 (Centered residual moments are not radially monotone). *Let $N = 2$ and $C = 1$. Write*

$$p_1 = \frac{1}{2} + \delta, \quad p_2 = \frac{1}{2} - \delta.$$

Then Q_1 equals p_2 with probability p_1 and p_1 with probability p_2 . A direct calculation gives

$$\text{Var}(Q_1) = \delta^2 - 4\delta^4.$$

This increases near $\delta = 0$ but decreases as δ approaches $1/2$. Thus centered moments of the residual mass are not governed by the radial transform order above.

8 Berthet's relation revisited

Berthet [3] observed that, for a partial coupon collection and an LRU cache under the same IRM popularity, the miss rate and the forward difference of the coupon waiting-time expectation are asymptotically related by

$$\text{MR}_{LRU}(C; p) (\mathbb{E}T_{C+1} - \mathbb{E}T_C) \approx 1.$$

The exact residual identities show that the exact product is

$$\text{MR}_{LRU}(C; p) (\mathbb{E}T_{C+1} - \mathbb{E}T_C) = \mathbb{E}Q_C \mathbb{E}Q_C^{-1}.$$

Thus Berthet's relation is exact at uniform popularity, where $Q_C = (N - C)/N$ is deterministic, and is approximately true whenever Q_C is concentrated.

The transform order proved here gives a sharper finite- N picture. Along every nonconstant ray away from uniform,

$$\mathbb{E}Q_C \downarrow, \quad \mathbb{E}Q_C^{-1} \uparrow,$$

and, more generally,

$$\mathbb{E}e^{-tQ_C} \uparrow, \quad \mathbb{E}Q_C^{-s} \uparrow, \quad \mathbb{E}Q_C^r \downarrow \quad (0 < r \leq 1).$$

LRU and coupon collecting are therefore linked not by a single approximate reciprocal formula but by an exact common residual-hazard law. LRU observes the residual mass through the linear cost q ; coupon collection observes the same residual mass through the reciprocal cost q^{-1} .

9 Relation to the positive pair-kernel LRU proof

The radial LRU theorem in [13] proves

$$\frac{d}{d\theta} \text{Hit}_C(p_\theta) > 0$$

by deriving an explicit formula of the form

$$\frac{d}{d\theta} \text{Hit}_C(p_\theta) = \frac{1}{N\theta} \sum_{a < b} (p_a(\theta) - p_b(\theta))^2 K_{ab}^{(C)}(p_\theta), \quad K_{ab}^{(C)}(p_\theta) > 0.$$

The present Gumbel proof gives a stronger conclusion but does not replace the value of that local formula. The two methods reveal different structures.

- (1) The Gumbel crossing proof gives a global monotone coupling of the whole Plackett–Luce ranking along a ray from uniform. It proves increasing-convex order for M_C and gives the LRU–coupon transfer results.
- (2) The pair-kernel proof gives an explicit finite-dimensional derivative certificate for the hit rate. It is algebraic, computable, and useful for sensitivity analysis. It also clarifies exactly why ray directions from uniform are special.
- (3) The occupancy-Jacobian proof in [13] explains why full Schur-convexity fails while radial monotonicity survives: the sign-indefinite local terms collapse to positive squares only in the radial direction.

Thus the new theorem subsumes the expectation-level conclusion of [13] and extends it to transform order, but the earlier proof remains valuable as a local analytic and algebraic certificate.

10 Further directions

The results suggest several natural extensions.

- (1) **Joint residual-mass order.** The present theorem controls each fixed Q_C separately. A stronger theorem would order the vector

$$(Q_0, Q_1, \dots, Q_{N-1})$$

or the vector of coupon-collector increments. Such a result could imply stochastic monotonicity of full partial collection times T_j , not merely their one-step increments.

- (2) **Other ranking paths.** The crossing argument applies whenever scores have the form $G_i + \eta_i(t)$ and the pairwise differences $\eta_i(t) - \eta_j(t)$ move monotonically according to a fixed item score. This includes softmax-temperature paths $p_i(\beta) \propto e^{\beta v_i}$, and suggests a broader monotone-comparative-static theory for Plackett–Luce rankings.
- (3) **Other caching policies.** The argument is special to LRU/MTF because the stationary order is Plackett–Luce. Policies with different stationary cache laws, such as CLIMB, would require different Gibbs, total-positivity, or polynomial methods. The positive-kernel and occupancy-Jacobian methods of [13] may be more adaptable there than the common-Gumbel crossing proof.
- (4) **Dispersion of the residual mass.** The product

$$\mathbb{E}Q_C \mathbb{E}Q_C^{-1}$$

is a Jensen dispersion gap. The present theorem gives opposite monotonicities for the two factors, but it does not assert monotonicity of the product. Understanding this gap more precisely would sharpen the interpretation of Berthet’s approximation.

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This paper was developed in response to the observation that the LRU miss rate and the coupon-collector discovery increment are exact positive and reciprocal moment transforms of the same size-biased residual mass.

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